

# Jialin Huang

## Software Engineer

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Open to relocation: Netherlands / Germany / EU · Mandarin (native), English (professional)

Fullstack engineer with a frontend foundation, eight years across six industries. The past year I built end to end on an enterprise security product: a multi-tenant policy console migrated to Remix, access-control APIs, and native browser code. I build because it is the craft I want for life.

### SKILLS

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**Frontend:** React, Next.js, Remix, Vue, Canvas, Egret, Tailwind CSS, Emotion, Redux, C#/.NET

**Backend:** NestJS, Apollo GraphQL, Golang, Gin, WebSocket, Node.js, GraphQL, SignalR, Firebase Auth, Python/Flask

**Security & Domain:** RBAC, SAML, OAuth, audit logging, Zero Trust access, multi-tenant SaaS

**Infrastructure:** PostgreSQL, Docker, Kubernetes, Jenkins, AWS Lambda, AWS Cognito, Google Analytics, Firebase, GCP

### EXPERIENCE

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#### MammothCyber

Senior Full Stack Engineer · Cybersecurity · Taipei (Hybrid) · Apr 2025 - Present

Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; security domain (access control, audit, policy config)

##### Console Migration to Remix

Migrated console pages onto Remix (React 18, Mantine, TanStack Query/Table, Vite) off an earlier Vue / Next.js stack, page by page with new and old running side by side

##### Policy Configuration Console

Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request

Built the rules and content-profile editor in Remix, type-safe end to end with reactive validation

##### Secure Browser & Extension (Chromium)

Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Built the login dialog in the companion Chrome extension (React / Redux), surfacing device info and recent failed logins

##### Identity, Access & Audit

Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients

Integrated Google / Apple social login into Cognito on top of the existing SAML, reconciling the OAuth scopes for external IdPs and reworking several Cognito Lambda triggers, including the PreTokenGeneration hook that maps each social user to its tenant

Integrated credentials and storage across AWS, Azure, and GCP (presigned URLs, Key Vault signing)

##### Platform & Build Infrastructure

Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0

Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree

Cut two Docker images ~80% with bundling and pnpm deploy --prod: Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB; compressed registry size ~540 MB → 89 MB (-84%) and 628 MB → 180 MB (-71%)

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## Bunastar

Frontend Engineer · Gaming · Taipei (Hybrid) · Nov 2024 - Apr 2025

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

### Southeast Asian Gaming Platform - Pok Deng

Card game logic: betting, dealing, scoring, payout

Real-time card sync over WebSocket

Canvas rendering engine, optimized with sprite sheets

Cross-domain communication via postMessage API

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## Confetti

Full Stack Engineer · Gaming · Taipei (Hybrid) · Dec 2022 - Oct 2024

### confetti.studio

Y2K landing page

### Cashing Platform

Built and iterated payment and cash-out flows across multiple product lines

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## SCMP

Frontend Engineer · Media · Hong Kong (Remote) · Oct 2021 - Nov 2022

South China Morning Post: high-traffic international news platform, delivered fully remote

### scmp.com

Improved Core Web Vitals (LCP, CLS), Google's SEO ranking signals, on scmp.com, a high-traffic international news site: fixed the homepage hero's layout shift, lazy-loaded below-fold images, and tuned SSR data prefetch

Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)

Built the site-wide tracking plan across sections: GA events through the GTM dataLayer for traffic, and Mixpanel via SDK keyed on the reader's uuid for per-user product behavior

Applied GDPR right-to-erasure in the comment system: surfaced a gdpr\_erased flag through the GraphQL layer so erased users' names render as Anonymous

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## Futuresai

Frontend Engineer · Finance · Taipei (Hybrid) · Nov 2020 - Oct 2021

### futures-ai.com

Built a subscription-billing system with multiple tiers

Live trading strategy articles with graphical analysis

Algorithm-driven financial insights

### Trading Dashboard + Admin + Auth

Monorepo rewrite with SignalR for real-time trading-data streaming

SMS and LINE push notifications

Firebase auth middleware with custom membership logic

*Real-time products in a financial-trading domain*

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## Dentalltw

Frontend Engineer · Healthcare · Taipei (Hybrid) · Jul 2018 - Nov 2020

Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market

### Dentall Clinic Management

Built the desktop clinic app (UWP, C#/.NET) for registration, appointments, billing, and prescriptions, integrated with Taiwan's NHI (National Health Insurance) system

Built Taiwan's first SVG-based dental chart for procedure documentation

### Vietnam Market Expansion

Built the Vietnam web app (React, Redux): the clinic software localized, with multi-language support and SMS

### CI/CD Pipeline

Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

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## SAVY

Frontend Intern · Taipei (Hybrid) · Apr 2018 - Jul 2018

### Community Help Board

A SPA for posting and browsing local community help requests

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## PROJECTS

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### Palimpsest · Angular, TypeScript, Bash, Claude Code

A boilerplate for giving an AI agent a governed self. The identity file models you, not a persona, and shell hooks enforce the rules the model cannot talk its way out of.

Three governance speeds in one charter: a bottomline no in-session pressure can move, principles that shift only on new evidence, and a free drift layer.

Enforcement lives outside the model — a shell hook refuses the write when text hits the blocklist, instead of asking the model to comply.

An Angular dashboard mapping 11 "AI soul" projects on two axes, built by cloning each repo and reading its identity files: 5 of the 11 enforce their constraints, the rest only declare them.

trials/ treats self-claims as hypotheses — predictions are hashed and sealed before the answers arrive, then scored against them.

### URL Shortener · Go, Gin, PostgreSQL, Redis, Kafka, ClickHouse

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect

Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta

Live click dashboard over Server-Sent Events: a pub/sub hub fans out each redirect to all clients, plus a per-link 2s stream merging the Postgres total with the Redis delta

Hand-rolled a Redis-bitset Bloom filter to short-circuit lookups for keys that do not exist

### **221b** · Python, ChromaDB, sentence-transformers, MCP

My outie and my innie in one searchable self. Severance, but merged instead of split. Ask what my brain has shipped, how I think, or "what would I do?", and it answers as me.

One corpus of my blog, my Claude sessions, and my inner config (soul, persona, style).

Chinese question → embeddings retrieve → reranker reorders → my English writing.

Answers "if you were me, how would you decide?" from my values and past calls.

### Game of Life · WebGL, TypeScript

Conway's Game of Life running on the GPU.

Simulated 770K+ cells at 60fps with a WebGL fragment-shader pipeline

### Innies · React, TypeScript

A playable Lumon MDR number-sorting game from Severance (人生切割術).

Recreated the macrodata-refinement interface as an interactive, playable piece

### Kubelens · Angular, TypeScript

If you have kubectl, you have Kubelens. Nothing to install.

Interactive graph of how Deployments, Pods, Services, and Ingresses connect.

A workspace modeled on macOS desktops: drag any resource into its own panel and run rollouts right there.

Snapshot the whole cluster to browse offline.

**National Chengchi University (NCCU)** · B.A. in Economics